

Berkay Özbek

PhD, Electrical and Electronic Engineering | Analog IC Design | Wireless Implantable Bioelectronics

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Research Profile

My research lies at the intersection of analog and mixed-signal integrated circuits, wireless microsystems, and bio-electronic medicine. I develop energy-efficient circuit and system architectures that enable miniaturised biomedical devices to operate wirelessly, securely, and reliably under stringent power, size, and communication constraints. My experience spans full-custom CMOS design, advanced power-management and RF interfaces, tape-out, hardware prototyping, and system-level validation across 180 nm, 65 nm, and 14 nm process nodes in both academic and industrial environments. As an independent investigator, I aim to establish a research programme on scalable wearable, implantable, and injectable bioelectronic platforms for minimally invasive diagnostics, closed-loop neuromodulation, and personalised therapies that improve human health and quality of life.

Education

Imperial College London, London, United Kingdom 2022–2026
PhD, Electrical and Electronic Engineering

- Supervisor: Prof. Timothy G. Constandinou
- Thesis: *A Self-Regulating RF Energy Harvesting and Secure Bidirectional Telemetry System for Distributed Neural Implants*

Middle East Technical University, Ankara, Turkey 2019–2021
MSc, Electrical and Electronics Engineering, Major in Microelectronics

- GPA: 3.86/4.00
- Thesis: *Low-Power Highly-Programmable Analog Front-End for 12-Channel Fully Implantable Cochlear Implants*
- Supervisor: Prof. Haluk Kulah

Middle East Technical University, Ankara, Turkey 2014–2019
BSc, Electrical and Electronics Engineering

Research and Professional Experience

Postdoctoral Research Associate, Imperial College London, London July 2026 – Present
• Will be developing low-power CMOS driver and control circuits for remotely addressable biohybrid neural interfaces, enabling precise spatiotemporal modulation of engineered neural cells.

Analog/PMU Design Intern, Qualcomm, Cambridge, United Kingdom February–August 2026
• Designing and verifying transistor-level analog and power-management circuits for a 14-nm CMOS tape-out, including schematic and post-layout analysis across PVT conditions.
• Contributing to design reviews, verification closure, and mixed-signal integration within an industrial chip-development flow.

PhD Researcher, Next Generation Neural Interfaces Lab, Imperial College London 2022–2026
• Developed NeuroStreamX, a measured 433-MHz, 65-nm, 1.01 mm² implant SoC integrating RF energy harvesting, regulation-aware power management, bidirectional telemetry, encryption, authentication, and hardware security for distributed neural implants.
• Implemented an adaptive hardware root-of-trust combining a dual-mode TRNG and autonomous OTP provisioning for implant credentials, achieving 168 nW–44.3 Mb/s, 3.95 pJ/bit, AIS31/NIST compliance, and 400 mV supply-injection robustness.
• Introduced a wide PDR ULP-assisted RF rectifier core that improves weak-input sensitivity while suppressing

strong-input reverse loss, achieving 91.5% PCE, -25.3 dBm sensitivity, and 24.2 dB PDR in 65 nm CMOS.

- Developed global-peak adaptive power management using absolute-peak impedance-matching-network tuning and a self-regulating rectifier-PMU, avoiding local-peak lock-in and headroom loss while achieving 85.12% RF-DC efficiency, -22.28 dBm sensitivity, and 32.48 dB dynamic range.
- Implemented encrypted secure single-carrier telemetry with low-modulation-index ASK downlink, LSK uplink, dual-loop CDR, encrypted packetization, TDMA access, and 64-bit ID + 64-bit password authentication, achieving 100 kb/s with $\text{BER} < 2 \times 10^{-5}$ at 34.5 μW .

Research Assistant, Fully Implantable MEMS-Based Autonomous Cochlear Implant Project, Middle East Technical University 2019–2021

- Designed a low-noise, low-power, and highly programmable analog front-end for non-resonant-mode piezoelectric acoustic transducers in fully implantable cochlear implants.
- Developed 12-channel signal-conditioning circuits for implantable acoustic sensing, with emphasis on tunability, low power consumption, and compatibility with different sensor responses.
- Completed one tape-out in 0.18 μm CMOS technology.
- Gained hands-on cleanroom experience in lithography, metallization, plasma processes, and etching.

Visiting Researcher, Micro and Nanosystems Research Group, KTH Royal Institute of Technology June–August 2018

- Worked on the Dried Blood Spot Sampling at Home in Therapeutic Drug Monitoring project, funded by the European Research Council.
- Designed a PCB-level electronic system for recording time stamps in a microfluidic blood sampling platform, improving reliability of sample interpretation.

Teaching & Supervision Experience

Teaching Experience

Graduate Teaching Assistant (Lead) — Full Custom IC Design (MSc/MEng), Imperial College London 2022–2025

- Coordinated laboratory components of this postgraduate module, with responsibility for scheduling, content development, and administration.
- Developed laboratory and lecture materials, including tutorials, assignments, coursework workshops, and examination questions.
- Coordinated, and mentored a team of 10 GTAs, overseeing consistent weekly lab delivery across all cohorts.
- Delivered office hours providing guidance on advanced CMOS theory, the full-custom IC design flow (schematic → simulation → layout), mixed-signal trade-offs, DRC/LVS verification, and industry-standard EDA toolchains.
- Managed the full assessment pipeline, including marking and handling student grade appeals.

Graduate Teaching Assistant (Lead) — Advanced Design of Integrated Circuits (ADIC) (MSc/MEng), Imperial College London 2022–2025

- Coordinated laboratory components of this advanced postgraduate module, serving as the academic point of contact for lab delivery and student assessment.
- Designed laboratory content and assessments, covering advanced analogue/mixed-signal and digital system design using Cadence Design Systems (CDS) on an AMS 0.35 μm process.
- Supervised a team of 5 GTAs, ensuring rigorous delivery of weekly sessions on: PSRR/CMRR simulation, noise and Monte Carlo analysis, PSS/PAC analysis, analogue layout, post-extraction simulation, and pad-ring/ESD design.
- Managed the full assessment pipeline, including marking and resolving student grade objections.

Graduate Teaching Assistant — Analysis and Design of Circuits (Year 1 UG), Imperial College London 2022–2025

- Delivered problem-solving classes and laboratory sessions for core first-year undergraduate EEE students, covering circuit analysis (Kirchhoff's laws, Thévenin/Norton), AC/DC analysis, small-signal transistor modelling (BJT/MOSFET), op-amp circuits, filter design, and SPICE simulation.

Graduate Teaching Assistant — Instrumentation (MSc/MEng), Imperial College London 2022–2025

- Supervised MSc/MEng students through coursework design projects involving sensor interface circuit design, PCB prototyping, embedded data acquisition, and measurement system calibration; assessed via oral examination and technical report.

Graduate Teaching Assistant — Mathematics 1A (Year 1 UG), Imperial College London 2022–2025

- Delivered problem-solving classes for first-year EEE undergraduates, covering calculus, complex numbers, ODEs, and engineering applications in signal processing and electromagnetism.

Student Supervision

Postgraduate & Undergraduate Research Supervisor, Imperial College London 2022–2024

- Supervised 2 MSc thesis students in advanced microelectronics research:
 - *Reconfigurable High-Efficiency All-PMOS Charge Pump* — student subsequently secured a PhD position at Nanyang Technological University (NTU) and published at PRIME 2024.
 - *Low-Power CMOS Temperature Sensor Design* — covering analogue IC design and low-power microelectronics.
- Supervised 1 undergraduate summer research student on *Programmable Pulse Generation for Characterisation of CMOS Antifuse Memory Cells*, spanning PCB design, instrumentation, and memory device characterisation.

Publications

Manuscripts in Preparation

1. **B. Özbek**, J. Ge and T. G. Constandinou, “A Self-Tuning 433-MHz RF Energy-Harvesting System for Wirelessly Powered Neural Implants With Absolute-Peak MPPT and ULP-Assisted Rectification,” *invited journal extension under preparation for submission to IEEE Transactions on Biomedical Circuits and Systems (TBioCAS)*, 2026.
2. **B. Özbek**, J. Ge, C. Bernardoni and T. G. Constandinou, “A Secure Multiple-Access Wireless Communication System for Distributed Neural Implants,” *manuscript under preparation for submission to Nature Communications for the Nature Portfolio Collection “Implantable Wireless Communication Technologies.”*

Journal Articles

1. **B. Özbek** and T. G. Constandinou, “A Closed-Loop System for Extracting Programming Parameters in Break-down Based OTP Memory,” *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, Early Access, 2026, doi: 10.1109/TVLSI.2026.3706590.
2. **B. Özbek** and T. G. Constandinou, “A 168 nW to 44.3 Mb/s Adaptable TRNG With 400 mV Attack-Resilient Hybrid RO Core,” *IEEE Solid-State Circuits Letters*, vol. 8, pp. 325–328, 2025, doi: 10.1109/LSSC.2025.3621413.

Conference Papers and Demonstrations

1. **B. Özbek** and T. G. Constandinou, “A 433-MHz Self-Regulated RF Energy Harvester With Absolute-Peak MPPT and ULP-Assisted Rectifier Core Achieving 91.5% PCE and 24.2-dB PDR,” *2026 IEEE Custom Integrated Circuits Conference (CICC)*, Seattle, WA, USA, 2026, pp. 1–5, doi: 10.1109/CICC65509.2026.11509465. *Invited for publication in IEEE Transactions on Biomedical Circuits and Systems (TBioCAS)*.
2. **B. Özbek** and T. G. Constandinou, “Live Demonstration: Pocket-Size USB–C TRNG With Interactive GUI for High-Entropy Security,” *2026 IEEE International Symposium on Circuits and Systems (ISCAS)*, Shanghai, China, 2026, p. 11, doi: 10.1109/ISCAS66217.2026.11562525.
3. J. Ge, V. Manatchinapisit, **B. Özbek**, and T. G. Constandinou, “Design and Validation of a Clocked gm-C Bandpower Extraction Circuit for Biosignal Analysis,” *2026 IEEE International Symposium on Circuits and Systems (ISCAS)*, Shanghai, China, 2026, pp. 2937–2941, doi: 10.1109/ISCAS66217.2026.11563007.
4. N. Martínez, **B. Özbek**, Y. Liu, D. Hacı, P. Feng, A. S. Idil, S. Ghoreishizadeh, N. Donaldson, P. Degenaar, A. Jackson, and T. G. Constandinou, “A SoC for an Active Implantable Microsystem for Closed-Loop Optogenetic Neuromodulation,” *2025 IEEE International Symposium on Circuits and Systems (ISCAS)*, London, United Kingdom, 2025, pp. 1–5, doi: 10.1109/ISCAS56072.2025.11043905.
5. **B. Özbek**, J. Ge, P. Feng, and T. G. Constandinou, “An Adaptive Impedance Matching System With Absolute Peak Detection for RF Energy Harvesting,” *2024 IEEE Biomedical Circuits and Systems Conference (BioCAS)*,

Xi'an, China, 2024, pp. 1–5, doi: 10.1109/BioCAS61083.2024.10798328.

6. J. Ge, **B. Özbek**, P. Feng, A. Mifsud, and T. G. Constandinou, “Flexible Electromagnetic-Lens for Wireless, Implantable Brain Machine Interfaces,” *2024 IEEE Biomedical Circuits and Systems Conference (BioCAS)*, Xi'an, China, 2024, pp. 1–5, doi: 10.1109/BioCAS61083.2024.10798285.
7. X. Chen, **B. Özbek**, P. Feng, and T. Constandinou, “A Reconfigurable High-Efficiency All-PMOS Charge Pump With Dual-Phase Clock Scheme,” *2024 19th Conference on PhD Research in Microelectronics and Electronics (PRIME)*, Larnaca, Cyprus, 2024, pp. 1–4, doi: 10.1109/PRIME61930.2024.10559669.
8. **B. Özbek** and T. G. Constandinou, “An Autonomous Zero-Mask Unique ID Generation System for Next-Generation Neural Interfaces,” *2023 IEEE Interregional NEWCAS Conference (NEWCAS)*, Edinburgh, United Kingdom, 2023, pp. 1–5, doi: 10.1109/NEWCAS57931.2023.10198170.
9. **B. Özbek** and H. Kùlah, “A 9.03 μ W Low Noise Highly Tunable Analog Front-End for Fully Implantable Cochlear Prosthesis,” *2022 IEEE Biomedical Circuits and Systems Conference (BioCAS)*, Taipei, Taiwan, 2022, pp. 349–353, doi: 10.1109/BioCAS54905.2022.9948682.
10. **B. Özbek** and H. Kùlah, “A 1-V Nanopower Highly Tunable Biquadratic Gm-CG_m-C Bandpass Filter for Fully Implantable Cochlear Implants,” *2021 IEEE Biomedical Circuits and Systems Conference (BioCAS)*, Berlin, Germany, 2021, pp. 1–5, doi: 10.1109/BioCAS49922.2021.9645031.

Grants, Fellowships, and Awards

International PhD Fellowship (approx. USD 120,000), The Scientific and Technological Research Council of Turkey (TUBITAK)	2022–2024
Individual Young Entrepreneur Grant (TRY 200,000), TUBITAK, Project No. 30831	2020
• Principal applicant for a project on next-generation self-powered cardiac pacemakers using MEMS-based piezoelectric energy harvesters from heartbeats to reduce battery replacement surgery. <i>Press coverage: METU News</i>	
Graduate Scholarship Program , The Scientific and Technological Research Council of Turkey (TUBITAK)	2019–2021
Outstanding Success Scholarship , Turkish Education Foundation (TEV)	2015–2019

Invited Talks and Media Engagements

Qualcomm Technology Forum , San Jose, CA, USA	Jun. 2026
Internet of Bodies Workshop , IEEE Custom Integrated Circuits Conference (CICC), Seattle, WA, USA	Apr. 2026
TRT World Roundtable , “What Can Brain Implants Do Now?”, London, UK	Feb. 2024

Academic and Professional Service

Head PhD Representative , Department of Electrical and Electronic Engineering, Imperial College London	2023–2025
• Represented PhD students at departmental level, coordinated communication between student representatives and academic staff, and contributed to improving the doctoral research environment.	
• Organized a department-wide Imperial EEE alumni event featuring eight invited speakers and attracting more than 100 students, researchers, and staff.	
Peer Reviewer , IEEE Journals and Conferences	2022–Present
• IEEE Transactions on Biomedical Circuits and Systems (TBioCAS), IEEE Open Journal of Circuits and Systems (OJCAS), IEEE International Symposium on Circuits and Systems (ISCAS), and IEEE Biomedical Circuits and Systems Conference (BioCAS).	
Executive Board Member , IEEE Student Branch, Middle East Technical University	2015–2017
• Initiated and co-organized the inaugural METU PES Congress, establishing an annual technical event that is scheduled to hold its 12th edition in 2027.	
• Contributed to student-led technical events, professional development activities, and IEEE community engagement.	

Technical Expertise

- **Analog and Power-Management IC Design:** Low-power analog and mixed-signal circuits, RF rectifiers, LDOs, bandgaps, charge pumps, voltage references, comparators, peak detectors, oscillators, AFEs, and self-regulating power-management circuits.
- **RF Energy Harvesting and Wireless Interfaces:** ULP RF rectifiers, adaptive impedance matching, inductive links, wireless power transfer, ASK/LSK telemetry, clock and data recovery, and robust power and data links for distributed neural implants.
- **Biomedical and Secure Integrated Systems:** Implantable neural and cochlear interfaces, low-noise biomedical AFEs, secure implant telemetry, TRNG-based security, antifuse OTP memories, hardware authentication, and multi-node communication architectures.
- **CMOS Design and Verification:** Tape-out experience in 180 nm, 65 nm, and 14 nm CMOS technologies; transistor-level design, PVT and Monte Carlo analysis, mixed-signal verification, layout review, parasitic extraction, post-layout simulation, and silicon bring-up.
- **EDA and Modelling Tools:** Cadence Virtuoso, Spectre, APS, Calibre DRC/LVS, HSPICE, LTspice, Verilog-A, SystemVerilog, MATLAB/Simulink, COMSOL, and Altium Designer.
- **Experimental Characterisation:** RF power and efficiency measurements, impedance and network analysis, spectrum analysis, oscilloscope-based debugging, PCB-level testing, signal-integrity evaluation, and characterisation of wireless biomedical ICs.

Professional Memberships

- Institute of Electrical and Electronics Engineers (IEEE)
- IEEE Circuits and Systems Society
- IEEE Solid-State Circuits Society

Languages

Turkish: Native

English: Fluent

References

- Prof. Timothy G. Constandinou, PhD Supervisor, Department of Electrical and Electronic Engineering, Imperial College London
- Assoc. Prof. Christos M. Papavassiliou, PhD Thesis Examiner, Department of Electrical and Electronic Engineering, Imperial College London
- Assoc. Prof. Shiwei Wang, PhD Thesis Examiner, School of Engineering, University of Edinburgh
- Manjunatha Poojary, Principal Engineer and Manager, Qualcomm, Cambridge
- Prof. Haluk Kulah, MSc Supervisor, Department of Electrical and Electronics Engineering, Middle East Technical University
- Prof. Niclas Roxhed, Research Supervisor, Division of Micro and Nanosystems, KTH Royal Institute of Technology